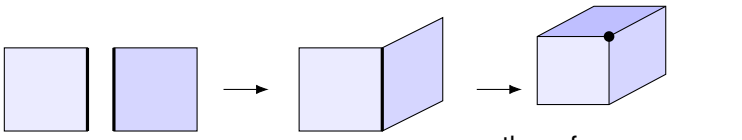
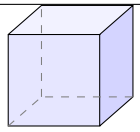
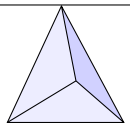
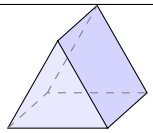
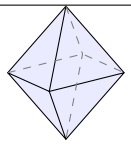
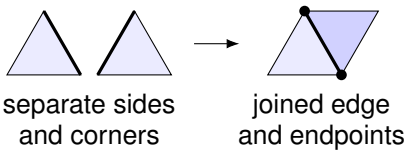
Use closed models with all their faces attached. A vertex is a corner of the assembled solid; an edge is shared by two faces. Count each assembled edge and vertex once. Tabs and glue overlaps do not count.



Problem 1: Find the vertices, edges and faces of each closed model.

Model	Vertices V	Edges E	Faces F
 cube			
 tetrahedron			
 triangular prism			
 octahedron			

Problem 2: Imagine cutting every face free. Find a rule that turns the number of separate face sides into the assembled edge count. Can the separate face-corner count alone tell you the vertex count?

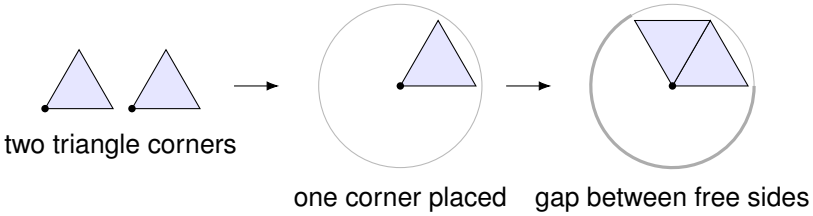


separate sides and corners

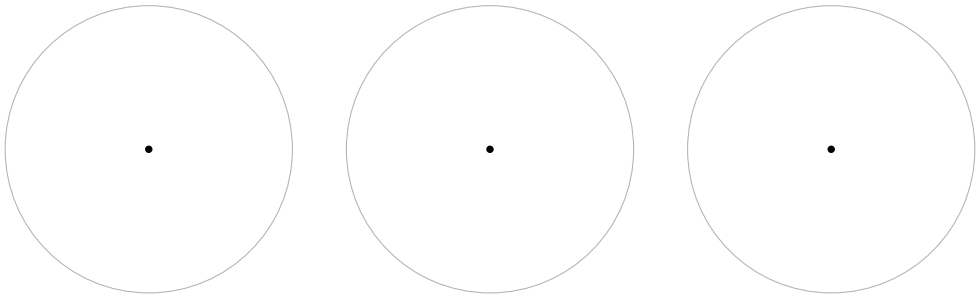
joined edge and endpoints

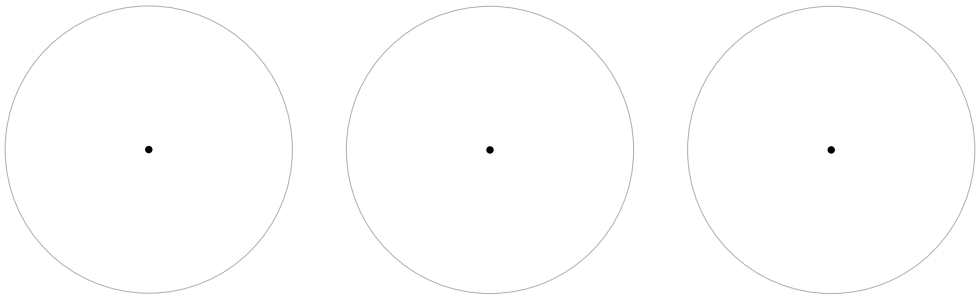
Faces before assembly	Separate sides	Assembled edges	Separate corners	Assembled vertices
6 squares: cube				
4 triangles: tetrahedron				
2 triangles and 3 squares: prism				
8 triangles: octahedron				

Join face corners around one point. Keep their sides meeting edge to edge and their marked vertices together. The gap is the missing part of a full turn when the corners lie flat.

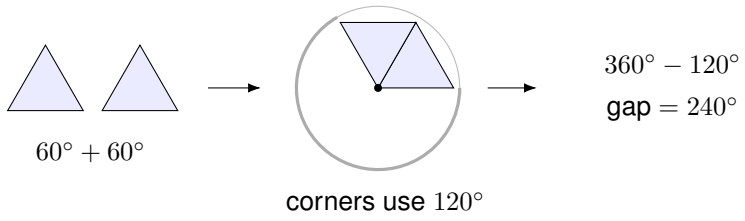


Problem 3: Which groups can close into a pointed corner with no inward dents? Which close flat with no gap? Make each group, then draw its gap with the corners laid flat.





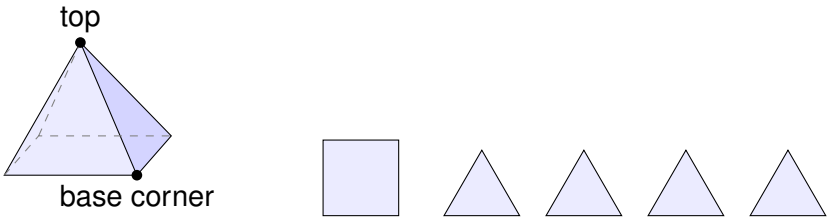
A full turn is $360°$. An equilateral triangle corner is $60°$; a square corner is $90°$. A vertex's gap is $360°$ minus its face-corner angles.



Problem 4: Find the gap at every vertex of each closed model. How many full turns do all its gaps make together?

Model	Vertices	Face corners at one vertex	Gap at one vertex	All gaps together
Cube				
Tetrahedron				
Triangular prism				
Octahedron				

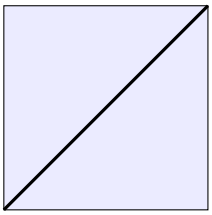
Problem 5: The square pyramid has one square face and four equilateral triangle faces. Which corner leaves the bigger gap, the top or a base corner? Find its total gap.



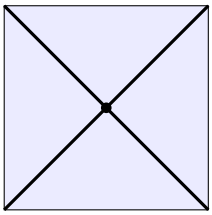
Vertex type	How many?	Face corners meeting there	Gap at each one
Top			
Base corner			

In a redrawn surface, marked points count as vertices, segments between vertices count as edges, and each face piece counts as a face.

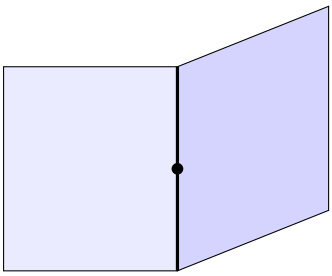
Problem 6: Compare $V - E + F$ for all four models in Problem 1. Keep the cube's shape and try each of these three changes on its own, starting from the original cube. What happens to $V - E + F$ and to the total gap? Can a new vertex have a gap while the cube keeps its shape?



one face divided by a diagonal



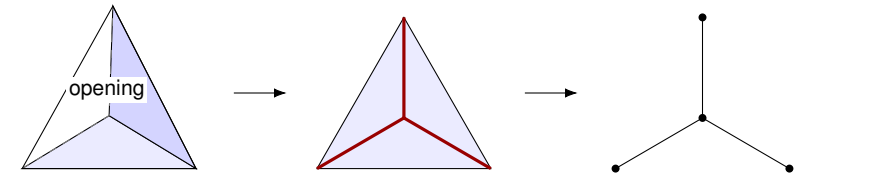
one face divided from its center



one vertex inserted along a shared edge

Closed surface	V	E	F	$V - E + F$	Total gap
Original cube					
One diagonal					
Center and 4 lines					
One edge vertex					

Convex means no dents. An opened drawing preserves the assembled edge and vertex connections. Count its bounded regions as faces; the outside is the removed face. A tree is a connected drawing with no closed loops.

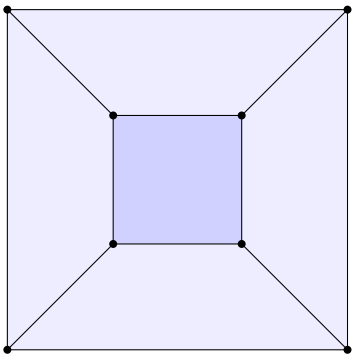


tetrahedron with one face removed

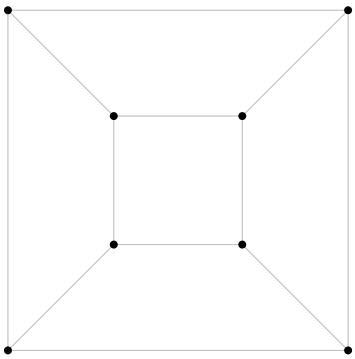
same three faces viewed through the opening

three boundary edges erased a tree remains

Problem 7: Explain why $V - E + F$ always has the same value for closed convex solids with polygon faces. Can an opened drawing be changed into a tree while keeping all its vertices connected?

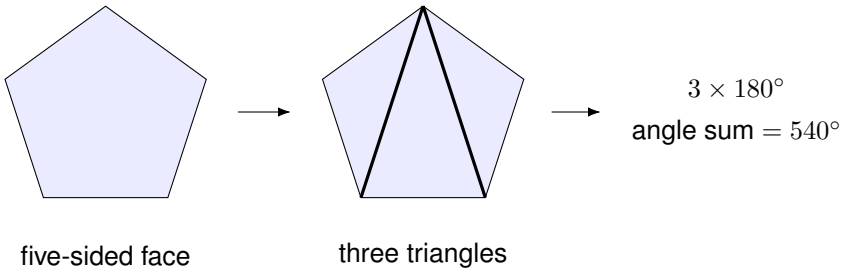


opened cube: five bounded faces



extra workspace

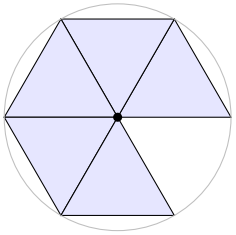
The angles of a triangle add to $180°$. A polygon with n sides can be divided into $n - 2$ triangles, so its corner angles add to $180(n - 2)°$.



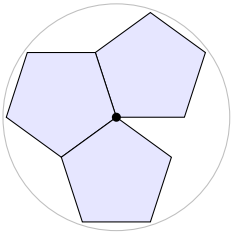
Problem 8: Explain why every closed convex solid with polygon faces has the same total gap, even when its vertex gaps are different.

A regular polygon has equal sides and equal angles. Every face of each solid below is one regular polygon.

Problem 9: Find the number of vertices, edges and faces each closed convex solid with polygon faces must have, without a model of the whole solid.



Triangle faces; five meet at every vertex



Pentagon faces; three meet at every vertex

Solid	Vertices	Edges	Faces
Five triangles at each vertex			
Three pentagons at each vertex			